SOUNDNESS AND RELATIVE COMPLETENESS OF THE FINITE-TRACE CALCULUS FOR RECURSIVE PROGRAMS

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Allowed standard prerequisites (editorial).

- Complete partial orders, least fixed points, fixed-point induction and Bekić decomposition. The full author semantic and proof appendices explicitly retain the source-local fixed-point arguments; general lattice fixed-point theorems remain standard background.

2. The Programming Language **Rec**

We define a simple programming language **Rec** with recursive procedures and give it a standard SOS semantics. We follow the notation of [26], adapted to **Rec** syntax.

Definition 2.1 (**Rec** Syntax)**.** A **Rec** program is a pair $\langle S, T\rangle$, where S is a statement with the abstract syntax defined by the grammar:

$$S \ ::= \ \textbf{skip} \mid x := a \mid S_1; S_2 \mid \textbf{if } b \textbf{ then } S_1 \textbf{ else } S_2 \mid m()$$

and where T is a procedure table given by $T \ ::= \ M^*$ with declarations of parameter-less procedures $M \ ::= \ m\,\{S\}$.

In the grammar, a ranges over arithmetic expressions **AExp**, and b over Boolean expressions **BExp**. Both are assumed to be side-effect free and are not specified further. All variables are global and range over the set of integers $\mathbb{Z}$. We assume programs are well-formed, i.e., only declared procedures are called, and procedure names are unique.

Example 2.2. Consider the **Rec** program $p_0 \stackrel{\mathsf{def}}{=} \langle S, T\rangle$ with statement $S \stackrel{\mathsf{def}}{=} x := 3; \mathit{even}()$ and procedure table:

$$T \quad \stackrel{\mathsf{def}}{=} \quad \begin{array}{l} \mathit{even}\ \{\textbf{if } x = 0 \textbf{ then } y := 1 \textbf{ else } x := x - 1; \mathit{odd}()\} \\ \mathit{odd}\ \ \{\textbf{if } x = 0 \textbf{ then } y := 0 \textbf{ else } x := x - 1; \mathit{even}()\} \end{array}$$

The intended semantics is that even terminates in a state where $y = 1$ if and only if it is started in a state where x is even and non-negative, and it terminates in a state where $y = 0$ if and only if it is started in a state where x is odd and non-negative.

Example 2.3. Consider the **Rec** program $p_1 \stackrel{\mathsf{def}}{=} \langle S, T\rangle$ with $S \stackrel{\mathsf{def}}{=} \mathit{down}()$ and procedure table:

$$T \quad \stackrel{\mathsf{def}}{=} \quad \mathit{down}\,\{\textbf{if } x > 0 \textbf{ then } x := x - 2; \mathit{down}() \textbf{ else skip}\}$$

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$$\text{Skip}\ \frac{-}{\langle \mathbf{skip}, s\rangle \Rightarrow s} \qquad \text{Assign}\ \frac{-}{\langle x := a, s\rangle \Rightarrow s[x \mapsto \mathcal{A}[\![a]\!](s)]}$$

$$\text{Seq-1}\ \frac{\langle S_1, s\rangle \Rightarrow s'}{\langle S_1; S_2, s\rangle \Rightarrow \langle S_2, s'\rangle} \qquad \text{Seq-2}\ \frac{\langle S_1, s\rangle \Rightarrow \langle S_1', s'\rangle}{\langle S_1; S_2, s\rangle \Rightarrow \langle S_1'; S_2, s'\rangle}$$

$$\text{If-1}\ \frac{-}{\langle \mathbf{if}\ b\ \mathbf{then}\ S_1\ \mathbf{else}\ S_2, s\rangle \Rightarrow \langle S_1, s\rangle}\ \text{if } \mathcal{B}[\![b]\!](s) = \mathbf{tt}$$

$$\text{If-2}\ \frac{-}{\langle \mathbf{if}\ b\ \mathbf{then}\ S_1\ \mathbf{else}\ S_2, s\rangle \Rightarrow \langle S_2, s\rangle}\ \text{if } \mathcal{B}[\![b]\!](s) = \mathbf{ff}$$

$$\text{Call}\ \frac{-}{\langle m(), s\rangle \Rightarrow \langle S, s\rangle}\ \text{if } m \text{ is declared as } m\,\{S\} \text{ in } T$$

Figure 1. SOS rules for **Rec**.

The intended semantics is that p_1 terminates in a state where $x = 0$ if and only if it is started in state where x is even and non-negative.

Remark 2.4. Loops can be defined with the help of (tail-)recursive programs. For example, a loop of the form "**while** b **do** S" can be simulated with a procedure declared in T as:

$$m\,\{\mathbf{if}\ b\ \mathbf{then}\ S; m()\ \mathbf{else\ skip}\}$$

using a unique name m and replacing the occurrence of the loop with a call to $m()$.

A standard, structural operational semantics (SOS) for **Rec** is defined in Figure 1 (sometimes referred to as small-step semantics). We use it as a base line when defining the denotational finite-trace semantics in Section 3.

Let Var be the set of program variables, and **State** the set of program states $s : Var \to \mathbb{Z}$. Let $\mathcal{A}[\![a]\!](s) \in \mathbb{Z}$ denote the integer value of the arithmetic expression a when evaluated in state s, and $\mathcal{B}[\![b]\!](s) \in \mathbb{T}$ denote the truth value of the Boolean expression b when evaluated in state s, both defined as expected.

A configuration is either a pair $\langle S, s\rangle$ consisting of a statement and a state, designating an initial or intermediate configuration; or a state s, designating a final configuration. To simplify notation we assume that S is evaluated relative to a **Rec** program with a procedure table T which is not explicitly specified.

The transitions of the SOS either relate two intermediate configurations, or an intermediate with a final one, and thus have the shape $\langle S, s\rangle \Rightarrow \langle S', s'\rangle$ or $\langle S, s\rangle \Rightarrow s'$, respectively.

Definition 2.5 (**Rec** SOS)**.** The structural operational semantics (SOS) of **Rec** is defined by the rules given in Figure 1.

The structural operational semantics of **Rec** induces a finite-trace semantics in terms of the sequences of states that are traversed from an initial to a final configuration when executing a given statement in the SOS. Let $\mathbf{State}^+$ denote the set of all non-empty, finite sequences of states. Formally, we define a function $\mathcal{S}_{sos}[\![S]\!] : \mathbf{Stm} \to 2^{\mathbf{State}^+}$, i.e., a function such that $\mathcal{S}_{sos}[\![S]\!] \subseteq \mathbf{State}^+$ for any statement S.

Definition 2.6 (Induced Finite-Trace Semantics)**.** Let S be a statement. Then, $\mathcal{S}_{sos}[\![S]\!]$ is defined as the set of (finite) sequences $s_0 \cdot s_1 \cdot \ldots \cdot s_n$ of

$$\mathcal{S}^0_{tr}[\![\textbf{skip}]\!]_\rho \stackrel{\text{def}}{=} \{s \cdot s \mid s \in \textbf{State}\} \qquad \mathcal{S}^0_{tr}[\![x := a]\!]_\rho \stackrel{\text{def}}{=} \{s \cdot s[x \mapsto \mathcal{A}[\![a]\!](s)] \mid s \in \textbf{State}\}$$
$$\mathcal{S}^0_{tr}[\![S_1; S_2]\!]_\rho \stackrel{\text{def}}{=} \mathcal{S}^0_{tr}[\![S_1]\!]_\rho \frown \mathcal{S}^0_{tr}[\![S_2]\!]_\rho \qquad \mathcal{S}^0_{tr}[\![m_i()]\!]_\rho \stackrel{\text{def}}{=} \rho(m_i)$$
$$\mathcal{S}^0_{tr}[\![\textbf{if } b \textbf{ then } S_1 \textbf{ else } S_2]\!]_\rho \stackrel{\text{def}}{=} (\sharp\mathcal{S}^0_{tr}[\![S_1]\!]_\rho)|_b \cup (\sharp\mathcal{S}^0_{tr}[\![S_2]\!]_\rho)|_{\neg b}$$

Figure 2. Finite-trace semantic equations for **Rec**.

states for which there are statements $S_0, S_1, \ldots, S_{n-1}$ such that $S_0 = S$, $\langle S_i, s_i\rangle \Rightarrow \langle S_{i+1}, s_{i+1}\rangle$ for all $0 \leq i \leq n-2$, and $\langle S_{n-1}, s_{n-1}\rangle \Rightarrow s_n$.

Observe that in Definition 2.6 any state $s_0 \in \textbf{State}$ can serve as the initial state of a finite trace. Next we design a "direct-style", denotational finite-trace semantics that conforms with the SOS, in the sense that it is equal to the finite-trace semantics induced by the SOS.

3. A Denotational Finite-Trace Semantics for **Rec**

We define the semantic function $\mathcal{S}_{tr} : \textbf{Stm} \to 2^{\textbf{State}^+}$ with the intention that $\mathcal{S}_{tr}[\![S]\!] = \mathcal{S}_{sos}[\![S]\!]$. Unlike $\mathcal{S}_{sos}$, however, $\mathcal{S}_{tr}$ is defined directly, without referring to other semantic rules as SOS does (hence the term "direct-style"). We define $\mathcal{S}_{tr}[\![S]\!]$ in the style of denotational semantics, compositionally, by induction on the structure of S, and through defining equations.

Let us define a unary restriction operator on trace sets, for any trace set A and Boolean expression $b \in \textbf{BExp}$, as follows: $A|_b \stackrel{\text{def}}{=} \{s \cdot \sigma \in A \mid \mathcal{B}[\![b]\!](s) = \textbf{tt}\}$. It filters out all traces in A whose first state does not satisfy b. Another unary operator on trace sets is defined as $\sharp A \stackrel{\text{def}}{=} \{s \cdot s \cdot \sigma \mid s \cdot \sigma \in A\}$ which duplicates the first state in each trace in A. Finally, let us define the binary operator on trace sets: $A \frown B \stackrel{\text{def}}{=} \{\sigma_A \cdot s \cdot \sigma_B \mid \sigma_A \cdot s \in A \wedge s \cdot \sigma_B \in B\}$ which concatenates traces from A with traces from B that agree on the last and first state, respectively, but without duplicating that state, see also [15].

Definition 3.1 (Denotational Finite-Trace Semantics of **Rec**)**.** Let $M = \{m_1, \ldots, m_n\}$ be the set of procedure names in **Rec** program $\langle S, T\rangle$, where every m_i is declared as $m_i\,\{S_i\}$ in T. We define a helper function $\mathcal{S}^0_{tr}[\![S]\!]_\rho$ that is relativized on an interpretation $\rho : M \to 2^{\textbf{State}^+}$ of the procedure names, inductively, by the equations given in Figure 2. The duplication of the initial states in the equation for the **if** statement is needed to remain faithful to the SOS of **Rec**, which allocates a small step for evaluation of the guard b. We then introduce a semantic function $H : \left(2^{\textbf{State}^+}\right)^n \to \left(2^{\textbf{State}^+}\right)^n$ defined as:

$$H(\rho) \stackrel{\text{def}}{=} (\sharp\mathcal{S}^0_{tr}[\![S_1]\!]_\rho, \ldots, \sharp\mathcal{S}^0_{tr}[\![S_n]\!]_\rho) \ .$$

Function H is monotonic and continuous in the CPO with bottom $\left(\left(2^{\textbf{State}^+}\right)^n, \sqsubseteq, \varnothing^n\right)$, where $\sqsubseteq$ denotes point-wise set inclusion. Hence, by the Knaster-Tarski Theorem, it has a least fixed-point. Let ρ_0 denote this least fixed-point. The denotational finite-trace semantics of statements S of **Rec** is defined relative to this interpretation as: $\mathcal{S}_{tr}[\![S]\!] \stackrel{\text{def}}{=} \mathcal{S}^0_{tr}[\![S]\!]_{\rho_0}$.

$$\begin{array}{rcl@{\qquad}rcl}
\|p\|_{\mathcal{V}} & \stackrel{\text{def}}{=} & \{s \cdot \sigma \mid s \models p\} & \|R\|_{\mathcal{V}} & \stackrel{\text{def}}{=} & \{s \cdot s' \mid R(s,s')\} \\
\|X\|_{\mathcal{V}} & \stackrel{\text{def}}{=} & \mathcal{V}(X) & \|\phi_1 \wedge \phi_2\|_{\mathcal{V}} & \stackrel{\text{def}}{=} & \|\phi_1\|_{\mathcal{V}} \cap \|\phi_2\|_{\mathcal{V}} \\
\|\phi_1 \vee \phi_2\|_{\mathcal{V}} & \stackrel{\text{def}}{=} & \|\phi_1\|_{\mathcal{V}} \cup \|\phi_2\|_{\mathcal{V}} & \left\|\phi_1 \frown \phi_2\right\|_{\mathcal{V}} & \stackrel{\text{def}}{=} & \|\phi_1\|_{\mathcal{V}} \frown \|\phi_2\|_{\mathcal{V}}
\end{array}$$

$$\|\mu X.\phi\|_{\mathcal{V}} \stackrel{\text{def}}{=} \bigcap \left\{ \gamma \subseteq \mathbf{State}^+ \;\middle|\; \|\phi\|_{\mathcal{V}[X \mapsto \gamma]} \subseteq \gamma \right\}$$

Figure 3. Finite-trace semantic equations for formulas.

The finite-trace semantics agrees with the SOS of **Rec**, in the sense that $\mathcal{S}_{tr}[\![S]\!]$ coincides with the finite-trace semantics $\mathcal{S}_{sos}[\![S]\!]$ induced by the SOS, as defined in Definition 2.6.

Theorem 3.2 (Correctness of Trace Semantics)**.** For all statements S of **Rec**, we have: $\mathcal{S}_{tr}[\![S]\!] = \mathcal{S}_{sos}[\![S]\!]$

4. A Logic over Finite Traces

Our trace logic can be seen as an Interval Temporal Logic [15] with μ-recursion [28], or alternatively, as a temporal μ-calculus with a binary temporal operator corresponding to the chop operation over sets of traces (see, e.g., [31] for a general introduction to μ-calculus).

4.1. Syntax and Semantics of the Logic.

The philosophy behind our logic is to have logical counterparts to the statements of the programming language in terms of their finite-trace semantics. For instance, we use binary relation symbols that correspond to the atomic statements, and a chop operator corresponding to sequential composition. This design choice helps to simplify proofs of the properties of the logic and the calculus.

Definition 4.1 (Logic Syntax)**.** The syntax of the logic of trace formulas is defined by the following grammar:

$$\phi \;::=\; p \mid R \mid X \mid \phi_1 \wedge \phi_2 \mid \phi_1 \vee \phi_2 \mid \phi_1 \frown \phi_2 \mid \mu X.\phi$$

where p ranges over state formulas not further specified here, but assumed to contain at least the Boolean expressions **BExp**, R ranges over binary relation symbols over states, and X over a set RVar of recursion variables.

Definition 4.2 (Logic Semantics)**.** The finite-trace semantics of a formula ϕ is defined as its denotation $\|\phi\|_{\mathcal{V}} \subseteq \mathbf{State}^+$, relativized on a valuation $\mathcal{V} : \mathsf{RVar} \to 2^{\mathbf{State}^+}$ of the recursion variables, inductively by the equations given in Figure 3, where in the last clause $\mathcal{V}[X \mapsto \gamma]$ denotes the updated valuation.

One can show that the transformers $\lambda\gamma.\, \|\phi\|_{\mathcal{V}[X \mapsto \gamma]}$ are monotonic functions in the complete lattice $(2^{\mathbf{State}^+}, \subseteq)$ and hence, by Tarski's fixed-point theorem for complete lattices [32], they have least and greatest fixed-points. In particular, the least fixed point is simultaneously also the least pre-fixed point, hence the defining equation for $\mu X.\phi$. And because it is a fixed point, we have the following result for unfolding fixed-point formulas.

Proposition 4.3 (Fixed-Point Unfolding)**.** Let $\mu X.\phi$ be a formula and $\mathcal{V}$ a valuation. Then: $\|\mu X.\phi\|_{\mathcal{V}} = \|\phi[\mu X.\phi/X]\|_{\mathcal{V}}$.

$$\mathsf{stf}(\overline{X}, \mathbf{skip}) \stackrel{\mathsf{def}}{=} Id \qquad \mathsf{stf}(\overline{X}, x := a) \stackrel{\mathsf{def}}{=} Sb^a_x \qquad \mathsf{stf}(\overline{X}, S_1;\, S_2) \stackrel{\mathsf{def}}{=} \mathsf{stf}(\overline{X}, S_1)^\frown \mathsf{stf}(\overline{X}, S_2)$$

$$\mathsf{stf}(\overline{X}, \mathbf{if}\ b\ \mathbf{then}\ S_1\ \mathbf{else}\ S_2) \stackrel{\mathsf{def}}{=} (b \wedge Id^\frown \mathsf{stf}(\overline{X}, S_1)) \vee (\neg b \wedge Id^\frown \mathsf{stf}(\overline{X}, S_2))$$

$$\mathsf{stf}(\overline{X}, m()) \stackrel{\mathsf{def}}{=} \begin{cases} Id^\frown \mu X_m.\, \mathsf{stf}(\overline{X} \cup \{m\}, S_m) & m \notin \overline{X},\ m\,\{S_m\} \in T \\ Id^\frown X_m & \text{otherwise} \end{cases}$$

Figure 4. Definition of strongest trace formula.

Our calculus is based on closed formulas of the logic. Observe that fixed-point unfolding preserves closedness. For closed formulas the valuation $\mathcal{V}$ is immaterial to the semantics $\|\phi\|_{\mathcal{V}}$. In this case, we often omit the subscript and simply write $\|\phi\|$.

4.2. **Binary Relations.** We instantiate the set Rel of binary relation symbols with two specific relations:

$$\begin{array}{rcl} Id(s, s') & \stackrel{\mathsf{def}}{\iff} & s' = s \\ Sb^a_x(s, s') & \stackrel{\mathsf{def}}{\iff} & s' = s[x \mapsto \mathcal{A}[\![a]\!]\,(s)] \end{array}$$

These two relations are used to model **skip** and assignment statements, respectively. The transitive closure R^+ of a binary relation R over states is easily defined as a recursive formula in our logic as $R^+ \stackrel{\mathsf{def}}{\iff} \mu X.\, (R \vee R^\frown X)$.

Example 4.4. For any arithmetic expression a let Dec_a be a binary relation symbol interpreted as follows: $Dec_a(s, s') \stackrel{\mathsf{def}}{\iff} \mathcal{A}[\![a]\!]\,(s') \leq \mathcal{A}[\![a]\!]\,(s)$. That is, the value of a does not increase between two consecutive states. With this symbol, the formula Dec_a^+ expresses the property that the value of a does not increase throughout the whole execution of a program.

Remark 4.5. It is conceivable to define arbitrary binary relations (p, q) over state formulas p, q, i.e., a pair of pre- and post-condition. Alternatively, one can view a binary relation as a TLA action [20], and use primed versions of the variables to refer to their values in the second state of a pair. We do not require this generality in our examples.

4.3. **Strongest Trace Formulas.** Since our program logic is able to characterize program traces, and not merely pre- and postconditions or intermediate assertions, it is possible to establish a close correspondence between programs and trace formulas. This correspondence is captured by the following—constructive—definition of the strongest trace formula $\mathsf{stf}(S)$ for a given program S, which characterizes all terminating traces of S.

For each procedure declaration $m\,\{S_m\}$ in T, we create a fixed-point formula, whenever m is called the first time. Subsequent calls to m result in a recursion variable. To achieve this, we parameterize the strongest trace formula function with the already created recursion variables $\overline{X}$. This parameter is initialized to $\varnothing$ and is ignored by all case definitions except the one for a recursive call.

Definition 4.6 (Strongest Trace Formula)**.** Let $\langle S, T\rangle$ be a **Rec** program. The strongest trace formula for S, denoted $\mathsf{stf}(S)$, is defined as $\mathsf{stf}(S) \stackrel{\mathsf{def}}{=} \mathsf{stf}(\varnothing, S)$, where $\mathsf{stf}(X, S)$ is defined inductively in Figure 4.

$$\text{Skip}\ \frac{-}{\Gamma \vdash \mathbf{skip} : Id} \qquad \text{Assign}\ \frac{-}{\Gamma \vdash x := a : Sb_x^a}$$

$$\text{Seq}\ \frac{\Gamma \vdash S_1 : \phi_1 \quad \Gamma \vdash S_2 : \phi_2}{\Gamma \vdash S_1; S_2 : \phi_1 \frown \phi_2}$$

$$\text{If}\ \frac{\Gamma \vdash \mathbf{skip}; S_1 : \neg b \vee \phi \quad \Gamma \vdash \mathbf{skip}; S_2 : b \vee \phi}{\Gamma \vdash \mathbf{if}\ b\ \mathbf{then}\ S_1\ \mathbf{else}\ S_2 : \phi}$$

$$\text{Unfold}\ \frac{\Gamma \vdash S : \phi[\mu X.\phi/X]}{\Gamma \vdash S : \mu X.\phi} \qquad \text{Cons}\ \frac{\Gamma \vdash S : \phi' \quad \phi' \models \phi}{\Gamma \vdash S : \phi}$$

$$\text{Call}\ \frac{Y_m : \phi_m \notin \Gamma \qquad m\,\{S_m\} \in T \qquad \Gamma, Y_m : \phi_m \vdash S_m[\mathbf{skip}; Y_m/m(), \mathbf{skip}; Y_{m_1}/m_1(), \ldots, \mathbf{skip}; Y_{m_n}/m_n()] : \phi_m}{\Gamma \vdash m() : Id \frown \phi_m}$$

Figure 5. The rules of the proof calculus.

Example 4.7. For the program $even()$ with the procedure table of Example 2.2, the strongest trace formula is:
$Id \frown \mu X_{even}.\Big(\big(x = 0 \wedge Id \frown Sb_y^1\big) \vee$
$\big(x \neq 0 \wedge Id \frown Sb_x^{x-1} \frown Id \frown \mu X_{odd}.((x = 0 \wedge Id \frown Sb_y^0) \vee (x \neq 0 \wedge Id \frown Sb_x^{x-1} \frown Id \frown X_{even}))\big)\Big)$
The binder for X_{odd} can be removed without changing the semantics.

Example 4.8. For the program in Example 2.3, the strongest trace formula is:
$\mathsf{stf}(S) = Id \frown \mu X_{down}.\ \big((x > 0 \wedge Id \frown Sb_x^{x-2} \frown Id \frown X_{down}) \vee (x \leq 0 \wedge Id \frown Id)\big)$

Theorem 4.9 (Characterisation of Strongest Trace Formula)**.** Let $\langle S, T\rangle$ be a program of **Rec**. Then the following holds: $\|\mathsf{stf}(\varnothing, S)\| = \mathcal{S}_{tr}[\![S]\!]$

5. A Proof Calculus

We present a proof calculus for our logic in the form of a Gentzen-style deductive proof system, which is compositional both in the statement and the formula.

5.1. **Definition of the Calculus.** To obtain a compositional proof rule for procedure calls, its shape will essentially embody the principle of fixed-point induction explained in the Appendix. For this we need to represent recursion variables in the **Rec** language, whose syntax is extended with a set SVar of statement variables, ranged over by Y. We add these as a new category of atomic statements to **Rec**.

To define the semantics of programs in the presence of statement variables, we relativize the finite-trace semantics $\mathcal{S}_{tr}[\![S]\!]_{\mathcal{I}}$ of statements S on interpretations $\mathcal{I} : \mathsf{SVar} \to 2^{\mathbf{State}^+}$ of the statement variables, lifted from $\mathcal{S}_{tr}[\![S]\!]$ in the canonical manner.

Definition 5.1 (Calculus Syntax)**.** Judgments are of the form $S : \phi$, where S is a **Rec** statement, possibly containing statement variables, and ϕ is a closed trace formula. The sequents of the calculus are of the shape $\Gamma \vdash S : \phi$, where Γ is a possibly empty set of judgments.

5.1.1. Rules. The calculus has exactly one rule for each kind of **Rec** statement, except for statement variables which do not occur in initial judgments to be proven, but are only created intermittently in proofs by the (Call) rule. All statement rules are compositional in the sense that only the statement S in focus without any context appears in the conclusion.

The statement rules and two selected logical rules of the calculus are shown in Figure 5. The remaining logical rules, in particular the axioms, are the standard Gentzen-style ones and are omitted.

To prove the judgment $S : \phi$ for a program $\langle S, T\rangle$, we prove the sequent $\vdash S : \phi$. All rules, except the (Call) rule, leave the antecedent Γ invariant.

We first explain the two logical rules. The (Unfold) rule is based on Proposition 4.3 and is used to unfold fixed-point formulas. The consequence rule (Cons) permits to strengthen the trace formula ϕ in the succedent, i.e., the specification of the program under verification. This is typically required to achieve a suitable syntactic form of ϕ, or to strengthen an inductive claim. The rule assumes the existence of an oracle for proving the logical entailment between trace formulas.

The (Skip) and (Assign) rules handle the atomic statements, using the two binary relation symbols defined in Section 4.2. The (Seq) rule is a rule for sequential composition. Observe that it is compositional in the sense that no intermediate state between S_1 and S_2 needs to be considered.

The (If) rule is a compositional rule for conditional statements. The trace formulas $\neg b \vee \phi$ in the left premise (and $b \vee \phi$ in the right one) might at first appear counter-intuitive. Formula $\neg b \vee \phi$ is read as follows: We need not consider program S_1 for any trace, where $\neg b$ holds in the beginning, because these traces relate to S_2; otherwise, ϕ must hold. A more intuitive notation would be $b \to \phi$, but we refrain from introducing implication in our logic.

Unsurprisingly, the (Call) rule is the most complex. We associate with each declaration of a method m in T a unique statement variable Y_m. The antecedent Γ contains judgments of the form $Y_m : \phi_m$. One can think of the Y_m as a generic continuation of any recursive call to m of which we know that it must conform to its contract ϕ_m. Once this conformance has been established, the fact is memorized in the antecedent Γ. Therefore, the (Call) rule is triggered only when a call to procedure $m()$ is encountered the first time. This is ensured by the condition $Y_m : \phi_m \notin \Gamma$. To avoid having to apply the call rule again to recursive calls of $m()$, all such calls in the body S_m are replaced with $\mathbf{skip}; Y_m$, where the **skip** models unfolding and Y_m is justified by the assumption in Γ. Likewise, any other $Y_{m_i} : \phi_{m_i} \in \Gamma$ triggers an analogous substitution. Now the procedure body $S_m[\ldots]$ in the premise contains at most procedure calls to m' that do not occur in Γ.

If a different judgment than $Id^\frown\phi_m$ is to be proven, then rule (Cons) must be applied before (Call) to achieve the required shape.

Example 5.2. We prove the judgment $even() : \mathsf{stf}(even())$ for the program from Example 2.2 in Figure 6. We abbreviate the fixed-point formula in $\mathsf{stf}(even())$ from Example 4.7 with ϕ_{even}, so that $\mathsf{stf}(even()) = Id^\frown\phi_{even}$, the

$$\text{Seq}\ \frac{\cdots \vdash \mathbf{skip} : Id \qquad \text{Cons}\ \dfrac{Y_{even} : \phi_{even}, Y_{odd} : \phi_{odd} \vdash Y_{even} : \phi_{even}}{Y_{even} : \phi_{even}, Y_{odd} : \phi_{odd} \vdash Y_{even} : \mu X_{even}.(\cdots \frown \phi_{odd})}}{Y_{even} : \phi_{even}, Y_{odd} : \phi_{odd} \vdash \mathbf{skip}; Y_{even} : Id \frown \mu X_{even}.(\cdots \frown \phi_{odd})}$$

$$\vdots$$

$$\text{Call}\ \frac{Y_{even} : \phi_{even}, Y_{odd} : \phi_{odd} \vdash S_{odd}[\mathbf{skip}; Y_{even}/even()] : \phi_{odd}}{Y_{even} : \phi_{even} \vdash odd() : \mathsf{stf}(odd())}$$

$$\text{Cons}\ \frac{}{Y_{even} : \phi_{even} \vdash odd() : Id \frown \mu X_{odd}.((x = 0 \wedge Id \frown Sb_y^0) \vee (x \neq 0 \wedge Id \frown Sb_x^{x-1} \frown Id \frown \phi_{even})}$$

$$\vdots$$

$$\text{Unfold}\ \frac{}{\text{Call}\ \dfrac{Y_{even} : \phi_{even} \vdash S_{even} : \phi'_{even}[\phi_{even}/X_{even}]}{\vdash even() : \mathsf{stf}(even())}}$$

Figure 6. Proof of $even() : \mathsf{stf}(even())$.

body of $even()$ with S_{even}, and similarly for $odd()$. Moreover, we abbreviate

$$\phi'_{even} = \big(x = 0 \wedge Id \frown Sb_y^1\big) \vee \big(x \neq 0 \wedge Id \frown Sb_x^{x-1} \frown Id \frown \mu X_{odd}.((x = 0 \wedge Id \frown Sb_y^0) \vee (x \neq 0 \wedge Id \frown Sb_x^{x-1} \frown Id \frown X_{even}))\big)$$

The proof starts with the call rule, followed by an unfold of the fixed point formula ϕ_{even}. We now proceed with the other statement rules simultaneously on S_{even} and the formula on the right until we encounter the call of $odd()$ in S_{even}. Here we would like to apply the call rule to $odd()$, but we do not have ϕ_{odd} on the right, because the unfolding of ϕ_{even} went "too deep". To avoid a lengthy derivation at this point, we use the fact that trace formula on the right is equivalent to ϕ_{odd}, and use the consequence rule to obtain it.

Now we descend into $odd()$, similarly as before; however, because the judgment $Y_{even} : \phi_{even}$ is present on the left, the call rule replaces $even()$ in S_{even} with $\mathbf{skip}; Y_{even} : \phi_{even}$. Finally, we encounter the statement variable Y_{even}, but again the fixed point formula on the right is "too deep". After a second transformation we close the proof with an axiom.

Remark 5.3. It is easy to derive a rule for loops from the (Call) rule using the encoding $m\,\{\mathbf{if}\ b\ \mathbf{then}\ S; m()\ \mathbf{else\ skip}\}$ given in Remark 2.4. In a pure loop program no unprocessed recursive call except $m()$ ever occurs in the body, so the Call) rule is applicable with $\Gamma = \varnothing$. Rule (Call) instantiated to m and a suitable ϕ_m gives $S_m[\mathbf{skip}; Y_m/m()] = \mathbf{if}\ b\ \mathbf{then}\ S; \mathbf{skip}; Y_m\ \mathbf{else\ skip}$ in the premise on the right, so its single premise becomes: $Y_m : \phi_m \vdash \mathbf{if}\ b\ \mathbf{then}\ S; \mathbf{skip}; Y_m\ \mathbf{else\ skip} : \phi_m$. Subsequent application of rule (If) yields the two premises $Y_m : \phi_m \vdash \mathbf{skip}; S; \mathbf{skip}; Y_m : \neg b \vee \phi_m$ and $Y_m : \phi_m \vdash \mathbf{skip}; \mathbf{skip} : b \vee \phi_m$.. In each premise is a spurious **skip** resulting from evaluating the method call which is only due to the encoding. In addition, the antecedent is not needed to prove the second premise and can be removed. After reordering and simplification, rule (While) is obtained as:

$$\text{While}\ \frac{\Gamma \vdash \mathbf{skip} : b \vee \phi \qquad \Gamma, Y : \phi \vdash \mathbf{skip}; S; Y : \neg b \vee \phi}{\Gamma \vdash \mathbf{while}\ b\ \mathbf{do}\ S : \phi}$$

where Γ contains judgments of the form $Y : \phi$ originating from while loops encountered previously. These are only needed in a proof in the presence of nested loops.

5.1.2. Semantics.

Definition 5.4 (Calculus Semantics)**.** A judgment $S : \phi$ is termed valid in $\mathcal{I}$, denoted $\models_{\mathcal{I}} S : \phi$, whenever $\mathcal{S}_{tr}[\![S]\!]_{\mathcal{I}} \subseteq \|\phi\|$. A sequent $\Gamma \vdash S : \phi$ is termed valid, denoted $\Gamma \models S : \phi$, if for every interpretation $\mathcal{I}$, $S : \phi$ is valid in $\mathcal{I}$, whenever all judgments in Γ are valid in $\mathcal{I}$.

It is not possible to prove a judgment for $m()$ (or any other statement) that is stronger than its strongest trace formula. In this sense, $\mathsf{stf}(m())$ can be seen as a contract for m, in fact the strongest possible contract. This is captured in the following result:

Corollary 5.5 (Strongest Trace Formula)**.** Let S be a statement not involving any statement variables. Then the strongest trace formula $\mathsf{stf}(S)$ of S entails any valid formula for S. That is, if $\models S : \phi$, then $\mathsf{stf}(S) \models \phi$.

Proof. The result follows directly from Theorem 4.9 in the Appendix and Definition 5.4. □

5.2. **Soundness and Relative Completeness of the Calculus.** Our proof system is sound, in the sense that it can only derive valid sequents.

Theorem 5.6 (Soundness)**.** The proof system is sound: every derivable sequent is valid.

Our proof system is complete, in the sense that every valid sequent can be derived, relative to an oracle, used by rule (Cons), that provides logical entailment between trace formulas.

By Theorem 4.9 and Definition 5.4 we know that every judgment of the shape $S : \mathsf{stf}(S)$ is valid. We next show that all such judgments are derivable in our proof system. Together with Corollary 5.5, we obtain completeness, with the help of the rule (Cons). By definition: $\mathsf{stf}(m()) = Id^{\frown}\mu X_m.\,\mathsf{stf}(\{m\}, S_m)$, where S_m is the body of m. We abbreviate $\phi_m = \mu X_m.\,\mathsf{stf}(\{m\}, S_m)$ and in the following use $\mathsf{stf}(m()) = Id^{\frown}\phi_m$ without mentioning it explicitly.

Theorem 5.7 (Existence of Canonical Proof)**.** Let $\langle S, T\rangle$ be a **Rec** program with n many method declarations in T, and let $\Gamma = \{Y_{m_1} : \phi_{m_1}, \ldots, Y_{m_n} : \phi_{m_n}\}$. Then, the judgment $\Gamma \vdash S[\mathbf{skip}; Y_{m_1}/m_1(), \ldots, \mathbf{skip}; Y_{m_n}/m_n()] : \mathsf{stf}(S)$ is derivable in our calculus.

Corollary 5.8 (Relative Completeness)**.** The proof system is relatively complete: for every **Rec** program S without statement variables and every closed formula ϕ, any valid judgment of the form $S : \phi$ is derivable in the proof system.

Proof. By Theorem 4.9 we know that $\mathsf{stf}(S) \models \phi$, so we can use rule (Cons) to obtain $S : \mathsf{stf}(S)$, which is derivable by Theorem 5.7. □

Compared to a typical completeness proof of first-order Dynamic Logic, where the invariant is constructed as equations over the Gödelized program in the loop, the argument is much simpler, because the inductive specification logic is sufficiently expressive to characterize recursive programs (and loops as a special case). First-order quantifiers are not

even necessary, so our logic is not first-order, even though it is obviously Turing-hard and thus undecidable.

6. From Trace Formulas to Programs

In Section 4.3 we showed that any **Rec** program S can be translated into a trace formula $\mathsf{stf}(S)$ that has the same semantics in terms of traces. Now we look at the other direction: Given a trace formula ϕ, can we construct a canonical program $\mathsf{can}(\phi)$ that has the same semantics in terms of traces? In general, this is not possible, as the following example shows:

Example 6.1. Consider the trace formula: $Sb_y^0 \frown \mu X.\, \big(Id \vee Sb_y^{y+1} \frown X\big)$. Its semantics are the traces that count y up from 0 to any finite number. It is not possible to model the non-deterministic choice in the fixed point formula directly in **Rec**, because the number of calls is unbounded.

There is a **Rec** program that produces exactly the same traces as the formula above, up to auxiliary variables, for example, $y := 0; m()$, where m is declared as: $m()\,\{\textbf{if}\ (y \leq x)\ \textbf{then}\ y := y+1; m()\ \textbf{else skip}\}$. However, to transform an arbitrary formula with unbounded non-determinism in the number of calls to an equivalent one with non-deterministic initialization, is difficult and not natural.

6.1. Rec Programs with Non-deterministic Choice.

To achieve a natural translation from the trace logic to canonical programs, it is easiest to introduce non-deterministic choice in the form of a statement **if** $*$ **then** S_1 **else** S_2. The extension of language **Rec** with the corresponding grammar rule is called **Rec***.

The SOS rules for non-deterministic choice are:

$$*\text{-}i \quad \frac{-}{\langle \textbf{if}\ * \ \textbf{then}\ S_1\ \textbf{else}\ S_2, s\rangle \Rightarrow \langle S_i, s\rangle} \qquad i \in \{1,2\}$$

The finite-trace semantics of non-deterministic choice is:

$$\mathcal{S}_{tr}[\![\textbf{if}\ * \ \textbf{then}\ S_1\ \textbf{else}\ S_2]\!] \stackrel{\mathsf{def}}{=} \sharp\mathcal{S}_{tr}[\![S_1]\!] \ \cup\ \sharp\mathcal{S}_{tr}[\![S_2]\!]$$

The extension of Theorem 3.2 for non-deterministic choice is completely straightforward. Likewise, the theory of strongest trace formulas is easy to extend:

$$\mathsf{stf}(\textbf{if}\ * \ \textbf{then}\ S_1\ \textbf{else}\ S_2) \stackrel{\mathsf{def}}{=} Id \frown \mathsf{stf}(S_1) \vee Id \frown \mathsf{stf}(S_2)$$

It is easy to adapt the proof of Theorem 4.9. The corresponding calculus rule is:

$$\text{If-}* \quad \frac{\Gamma \vdash \textbf{skip}; S_1 : \phi \quad \Gamma \vdash \textbf{skip}; S_2 : \phi}{\Gamma \vdash \textbf{if}\ * \ \textbf{then}\ S_1\ \textbf{else}\ S_2 : \phi}$$

It is also easy to extend the proofs of Theorem 5.6 and Theorem 5.7. A more problematic aspect of the translation to canonical programs concerns formulas of the form $\phi_1 \wedge \phi_2$, because there is no natural programming construct that computes the intersection of traces. But in Definition 4.6 general conjunction is not required, so without affecting the results in previous sections we can restrict the syntax of trace formulas in Definition 4.1 to $p \wedge \phi$.

A final issue are the programs that characterize a trace formula of the form p with semantics $\|p\| = \textbf{State}^+|_p$. A program that produces such traces requires a **havoc** statement that resets all variables to an arbitrary value. This goes beyond non-deterministic choice quite a bit, but luckily, it is not required: As seen above, trace formulas of the form

p occur only as subformulas of $p \wedge \phi$. Further, formulas of the form $p \vee \phi$ occur only as intermediate formulas in derivations, and nowhere else. Altogether, for the purpose of mapping formulas to programs, we can leave out the production for p. The grammar in Definition 4.1 is thus simplified to:

$$\phi \ ::= \ Id \mid Sb^a_x \mid X \mid p \wedge \phi \mid \phi \vee \psi \mid \phi^\frown\psi \mid \mu X.\phi$$

In addition, we assume without loss of generality that all recursion variables in a trace formula have unique names.

6.2. **Canonical Programs.**

Definition 6.2 (Canonical Program)**.** Let ϕ be trace formula. The canonical program for ϕ, denoted $\mathsf{can}(\phi) = \langle S_\phi, T_\phi\rangle$, is inductively defined as follows:

$$\mathsf{can}(Id) \stackrel{\mathsf{def}}{=} \langle \mathbf{skip}, \epsilon\rangle \qquad \mathsf{can}(p \wedge \phi) \stackrel{\mathsf{def}}{=} \langle \mathbf{if}\ p\ \mathbf{then}\ S_\phi\ \mathbf{else\ diverge}, T_\phi\rangle$$

$$\mathsf{can}(Sb^a_x) \stackrel{\mathsf{def}}{=} \langle x := a, \epsilon\rangle \qquad \mathsf{can}(\phi \vee \psi) \stackrel{\mathsf{def}}{=} \langle \mathbf{if}\ *\ \mathbf{then}\ S_\phi\ \mathbf{else}\ S_\psi, T_\phi\, T_\psi\rangle$$

$$\mathsf{can}(\phi^\frown\psi) \stackrel{\mathsf{def}}{=} \langle S_\phi; S_\psi, T_\phi\, T_\psi\rangle \qquad \mathsf{can}(\mu X.\phi) \stackrel{\mathsf{def}}{=} \langle m_X(), T_\phi\,\{m_X\{S_\phi\}\}\rangle$$

$$\mathsf{can}(X) \stackrel{\mathsf{def}}{=} \langle m_X(), \epsilon\rangle$$

The definition contains the statement **diverge**. It is definable in the **Rec** language as **diverge** $\stackrel{\mathsf{def}}{=} abort()$, with the declaration $abort\,\{abort()\}$. We assume that any table T_ϕ contains the declaration of procedure $abort()$ when needed.

Example 6.3. We translate the formula in Example 6.1:

$$\mathsf{can}(Sb^0_y{}^\frown\mu X.(Id \vee Sb^{y+1}_y{}^\frown X)) = \langle y := 0; m_X(),\, T\rangle$$

where $T = m_X\,\{\mathbf{if}\ *\ \mathbf{then\ skip\ else}\ y := y + 1; m_X()\}$.

Proposition 6.4. We have $\mathcal{S}_{tr}[\![\mathbf{diverge}]\!] = \varnothing$.

Proposition 6.5. Let ϕ be an open trace formula, let T'_ϕ be declarations of its unbound recursion variables, and let $\mathsf{can}(\phi) = \langle S_\phi, T_\phi\rangle$. Then $\langle S_\phi, T_\phi\, T'_\phi\rangle$ is a well-defined **Rec*** program.

Evaluation of procedure calls and Boolean guards introduce stuttering steps as compared to the corresponding logical operators of least fixed-point recursion and disjunction, respectively. Hence, canonical programs obtained from a formula ϕ are statements, whose trace semantics is equal to the one of ϕ, but modulo stuttering: The two trace sets are equal when abstracting away the stuttering steps. Further, we say that statement S_1 refines statement S_2, written $S_1 \preceq S_2$, when $\mathcal{S}_{tr}[\![S_1]\!] \subseteq \mathcal{S}_{tr}[\![S_2]\!]$.

Definition 6.6 (Stuttering Equivalence)**.** Let $\tilde\sigma$ be the stutter-free version of a trace σ, i.e., where any subtrace of the form $s \cdot s \cdots s$ has been replaced with s. Define $\widetilde{A} = \{\tilde\sigma \mid \sigma \in A\}$. We say two trace sets A, B are stutter-equivalent, written $A \mathrel{\widetilde{=}} B$, if $\widetilde{A} = \widetilde{B}$.

It is easy to see that $A = B$ implies $A \mathrel{\widetilde{=}} B$. Let $\mathrel{\widetilde{\models}}$ and $\mathrel{\widetilde{\preceq}}$ denote entailment between formulas and refinement between statements, respectively, both modulo stuttering equivalence.

Unsurprisingly, the characterization of canonical programs resembles the one of strongest trace formulas, however, modulo stuttering equivalence.

Theorem 6.7 (Characterisation of Canonical Program)**.** Let ϕ be a closed trace formula, and let $\mathsf{can}(\phi) = \langle S_\phi, T_\phi \rangle$. Then $\mathcal{S}_{tr}[\![S_\phi]\!] \mathrel{\widetilde{=}} \|\phi\|$.

Finally, we can establish that $\mathsf{stf}(\cdot)$ and $\mathsf{can}(\cdot)$ form a Galois connection w.r.t. the partial orders $\widetilde{\models}$ on formulas and $\widetilde{\preceq}$ on statements.

Corollary 6.8. Let ϕ be a closed trace formula, and let $\mathsf{can}(\phi) = \langle S_\phi, T_\phi \rangle$. Then, for every statement S, we have: $\mathsf{stf}(S) \mathrel{\widetilde{\models}} \phi \quad \text{iff} \quad S \mathrel{\widetilde{\preceq}} S_\phi$.

Proof. By using Theorem 4.9 and Theorem 6.7. □

Appendix A. Proofs and More Examples

A.1. **Semantics.**

Example A.1. Let us execute the statement $\textbf{skip}; x := x - 1$ (with empty procedure table) from some arbitrary initial state s in the above SOS. That is, let us apply the rules to derive all configurations reachable from the initial one, which is $\langle \textbf{skip}; x := x - 1, s \rangle$. First, by means of the rules Skip and Seq-1, we derive the transition $\langle \textbf{skip}; x := x - 1, s \rangle \Rightarrow \langle x := x - 1, s \rangle$. Then, by applying the Assign rule, we derive the transition $\langle x := x - 1, s \rangle \Rightarrow s[x \mapsto s(x) - 1]$. We reached a final configuration, and thus execution terminates. The program execution (or run) we obtain is: $\langle \textbf{skip}; x := x - 1, s \rangle \Rightarrow \langle x := x - 1, s \rangle \Rightarrow s[x \mapsto s(x) - 1]$

Example A.2. We execute the statement $x := 2; \mathit{down}()$ with the procedure table from Example 2.3. The program execution (or run) we obtain from an arbitrary state s is:

$\langle x := 2; \mathit{down}(), s \rangle \Rightarrow \langle \mathit{down}(), s[x \mapsto 2] \rangle \Rightarrow$
$\langle \textbf{if } x > 0 \textbf{ then } x := x - 2; \mathit{down}() \textbf{ else skip}, s[x \mapsto 2] \rangle \Rightarrow$
$\langle x := x - 2; \mathit{down}(), s[x \mapsto 2] \rangle \Rightarrow \langle \mathit{down}(), s[x \mapsto 0] \rangle \Rightarrow$
$\langle \textbf{if } x > 0 \textbf{ then } x := x - 2; \mathit{down}() \textbf{ else skip}, s[x \mapsto 0] \rangle \Rightarrow \langle \textbf{skip}, s[x \mapsto 0] \rangle \Rightarrow s[x \mapsto 0]$

Example A.3. Building on Example A.1, the induced finite-trace semantics of the statement $\textbf{skip}; x := x - 1$ is the set $\{s \cdot s \cdot s[x \mapsto s(x) - 1] \mid s \in \textbf{State}\}$ of finite traces, all of which are of length 3.

We state some useful algebraic laws for finite-trace sets which we use later in our proofs:

$$A^\frown(B \cup C) = A^\frown B \cup A^\frown C \qquad (A \cup B)^\frown C = A^\frown C \cup B^\frown C$$
$$A \subseteq B \Rightarrow A^\frown C \subseteq B^\frown C \qquad (A^\frown B)|_b = A|_b{}^\frown B \qquad \sharp(A^\frown B) = \sharp A^\frown B$$

The following "unfolding" equivalence for procedure calls holds:

Proposition A.4. Let $m\,\{S\}$ be a method declaration. Then the following semantic equality holds: $\mathcal{S}_{tr}[\![m()]\!] = \sharp \mathcal{S}_{tr}[\![S]\!]$

Proof. We have:

$$\begin{array}{rcll} \mathcal{S}_{tr}[\![m()]\!] & = & \mathcal{S}^0_{tr}[\![m()]\!]_{\rho_0} & \{\text{By the definition of } \mathcal{S}_{tr}[\![S]\!]\} \\ & = & \rho_0(m) & \{\text{By the definition of } \mathcal{S}^0_{tr}[\![S]\!]_\rho\} \\ & = & \sharp \mathcal{S}^0_{tr}[\![S]\!]_{\rho_0} & \{\text{Since } \rho_0 \text{ is a fixed-point of } H\} \\ & = & \sharp \mathcal{S}_{tr}[\![S]\!] & \{\text{By the definition of } \mathcal{S}_{tr}[\![S]\!]\} \end{array}$$

□

Fixed-Point Induction and Bekič's Principle. It can be shown that $\rho_0 = \bigsqcup_{k \geq 0} H^k(\varnothing^n)$, where $\sqcup$ is point-wise set union, a result originally due to Kleene, but popularly known as the Knaster-Tarski Fixed-Point

Theorem. The interpretations $\rho^k \stackrel{\text{def}}{=} H^k(\varnothing^n)$ are referred to as fixed-point approximants of H.

Then, for a given transformer H and set A, a result of the type $LFP\, H \sqsubseteq A$ can be established by proving $H^k(\varnothing) \sqsubseteq A$ for all k, by mathematical induction on k. Since $\bigsqcup_{k\geq 0} H^k(\varnothing^n)$ is the least upper bound of the set of approximants, we obtain $LFP\, H \sqsubseteq A$. But sometimes one can prove a stronger result, namely that $\gamma \sqsubseteq A$ implies $H(\gamma) \sqsubseteq A$ for all γ. This immediately entails the above proof by mathematical induction—a proof schema commonly referred to as Fixed-Point Induction. We make use of these proof principles below.

We will also make use of another principle, known as Bekič's Principle, which allows a simultaneous fixed-point, such as ρ_0, to be expressed as a series of individual least fixed-points [4]. Using this principle, we can apply the above Fixed-Point Induction Principle, but on one procedure name at a time.

Presentation of Proofs. While the following inductive proofs over the statement structure were done for all cases, for sake of conciseness, we reproduce only the cases for conditional and method call: All other cases are straightforward and carry no additional information.

Proof of Theorem 3.2. Typically, such results are proved separately in the two directions, utilizing different induction principles.

(1) Proof of $\mathcal{S}_{tr}[\![S]\!] \subseteq \mathcal{S}_{sos}[\![S]\!]$.

For $\sigma \in \mathbf{State}^+$, $\sigma \in \mathcal{S}_{tr}[\![S]\!]$ is equivalent by definition to $\sigma \in \mathcal{S}^0_{tr}[\![S]\!]_{\rho_0}$, which in turn is equivalent to $\exists k \geq 0.\, \sigma \in \mathcal{S}^0_{tr}[\![S]\!]_{\rho^k}$. Hence, what we want to show is logically equivalent to showing that $\mathcal{S}^0_{tr}[\![S]\!]_{\rho^k} \subseteq \mathcal{S}_{sos}[\![S]\!]$ for all $k \geq 0$. We show this by mathematical induction on k.

The base case of $k = 0$ holds vacuously. For the induction case, assume $\mathcal{S}^0_{tr}[\![S]\!]_{\rho^k} \subseteq \mathcal{S}_{sos}[\![S]\!]$ for an arbitrary $k \geq 0$. We proceed by (an inner) induction on the structure of S. We proved all cases, but for lack of space, for this and all results that follow, we reproduce only the most interesting cases.

Case $S = \textbf{if}\ b\ \textbf{then}\ S_1\ \textbf{else}\ S_2$**.** Assume the result holds for S_1 and S_2. Let $s_0 \cdot s_1 \cdot \ldots \cdot s_n \in \mathcal{S}_{tr}[\![\textbf{if}\ b\ \textbf{then}\ S_1\ \textbf{else}\ S_2]\!]$. Then, by Definition 3.1, $s_0 = s_1$, and either $\mathcal{B}[\![b]\!](s_0) = \textbf{tt}$ and $s_1 \cdot \ldots \cdot s_n \in \mathcal{S}_{tr}[\![S_1]\!]$, or else $\mathcal{B}[\![b]\!](s_0) = \textbf{ff}$ and $s_1 \cdot \ldots \cdot s_n \in \mathcal{S}_{tr}[\![S_2]\!]$. By the induction hypothesis, then either $\mathcal{B}[\![b]\!](s_0) = \textbf{tt}$ and $s_1 \cdot \ldots \cdot s_n \in \mathcal{S}_{sos}[\![S_1]\!]$, or else $\mathcal{B}[\![b]\!](s_0) = \textbf{ff}$ and $s_1 \cdot \ldots \cdot s_n \in \mathcal{S}_{sos}[\![S_2]\!]$. By Definition 2.6, and by applying rule If-1 or If-2, respectively, we obtain that $s_0 \cdot s_1 \cdot \ldots \cdot s_n \in \mathcal{S}_{sos}[\![\textbf{if}\ b\ \textbf{then}\ S_1\ \textbf{else}\ S_2]\!]$.

Case $S = m()$ Let m be declared as $m\,\{S'\}$. Let $s_0 \cdot s_1 \cdot \ldots \cdot s_n \in \mathcal{S}^0_{tr}[\![m()]\!]_{\rho^{k+1}}$. Then, by the definition of $\mathcal{S}_{tr}[\![S]\!]$, we have $s_0 \cdot s_1 \cdot \ldots \cdot s_n \in \rho^{k+1}(m)$. But $\rho^{k+1} = H(\rho^k)$, and therefore $s_0 \cdot s_1 \cdot \ldots \cdot s_n \in \sharp\mathcal{S}^0_{tr}[\![S']\!]_{\rho^k}$, by definition of H. Hence, $s_0 = s_1$ and $s_1 \cdot s_2 \cdot \ldots \cdot s_n \in \mathcal{S}^0_{tr}[\![S']\!]_{\rho^k}$. By the outer induction hypothesis, $s_1 \cdot s_2 \cdot \ldots \cdot s_n \in \mathcal{S}_{sos}[\![S']\!]$, hence $s_0 \cdot s_1 \cdot \ldots \cdot s_n \in \sharp\mathcal{S}_{sos}[\![S']\!]$. By Rule Call and Definition 2.6, we finally obtain that $s_0 \cdot s_1 \cdot \ldots \cdot s_n \in \mathcal{S}_{sos}[\![m()]\!]$.

(2) Proof of $\mathcal{S}_{sos}[\![S]\!] \subseteq \mathcal{S}_{tr}[\![S]\!]$.

The proof proceeds by (strong) induction on the length of traces. Assume the result holds for all traces of length up to n, for all statements S. Let S be a statement and $s_0 \cdot \ldots \cdot s_n \in \mathcal{S}_{sos}[\![S]\!]$ be a trace of length $n+1$. Then, by Definition 2.6, there are statements $S_0, S_1, \ldots, S_{n-1}$ such that $S_0 = S$, $\langle S_i, s_i\rangle \Rightarrow \langle S_{i+1}, s_{i+1}\rangle$ for all $0 \leq i \leq n-2$, and $\langle S_{n-1}, s_{n-1}\rangle \Rightarrow s_n$. The proof that $s_0 \cdot \ldots \cdot s_n \in \mathcal{S}_{tr}[\![S]\!]$ proceeds by case analysis on the SOS rule

applied to obtain the first transition of the SOS trace. Again, we only show the most interesting cases.
Rule If-1. Then $S = \textbf{if}\ b\ \textbf{then}\ S_1\ \textbf{else}\ S_2$, $s_0 = s_1$ and $\mathcal{B}[\![b]\!](s_0) = \textbf{tt}$. By Definition 2.6, $s_1 \cdot \ldots \cdot s_n \in \mathcal{S}_{sos}[\![S_1]\!]$, and by the induction hypothesis, $s_1 \cdot \ldots \cdot s_n \in \mathcal{S}_{tr}[\![S_1]\!]$. Since $s_0 = s_1$ and $\mathcal{B}[\![b]\!](s_0) = \textbf{tt}$, we have $s_0 \cdot s_1 \cdot \ldots \cdot s_n \in (\sharp \mathcal{S}_{tr}[\![S_1]\!])|_b$. Then, by Definition 3.1, $s_0 \cdot s_1 \cdot \ldots \cdot s_n \in \mathcal{S}_{tr}[\![\textbf{if}\ b\ \textbf{then}\ S_1\ \textbf{else}\ S_2]\!] = \mathcal{S}_{tr}[\![S]\!]$.
Rule Call. Then $S = m()$ for a procedure m declared as $m\,\{S'\}$ for some statement S', and $s_0 = s_1$. By Definition 2.6, $s_1 \cdot \ldots \cdot s_n \in \mathcal{S}_{sos}[\![S']\!]$, and then, by the induction hypothesis, $s_1 \cdot \ldots \cdot s_n \in \mathcal{S}_{tr}[\![S']\!]$. Now, since $s_0 = s_1$, $s_0 \cdot s_1 \cdot \ldots \cdot s_n \in \sharp \mathcal{S}_{tr}[\![S']\!]$ and hence, by Proposition A.4, $s_0 \cdot s_1 \cdot \ldots \cdot s_n \in \mathcal{S}_{tr}[\![m()]\!] = \mathcal{S}_{tr}[\![S]\!]$.

□

Fixed-Point Induction. As in the domain of finite traces, one can also apply fixed-point induction in proofs about recursive formulas. In particular, a result of shape $\|\mu X.\phi\|_{\mathcal{V}} \subseteq A$ can be established by proving that $\gamma \subseteq A$ entails $\|\phi\|_{\mathcal{V}[X \mapsto \gamma]} \subseteq A$.

Proposition A.5 (Restricted Traces and State Formulas)**.** Let b be a Boolean expression, and ϕ a trace formula. Then $(\sharp \|\phi\|)|_b = \left\|b \wedge Id^{\frown}\phi\right\|$.

Proof. We have:

$$\begin{array}{lll} & (\sharp \|\phi\|)|_b & \\ = & \{s \cdot \sigma \in \sharp \|\phi\| \mid \mathcal{B}[\![b]\!](s) = \textbf{tt}\} & \\ = & \{s \cdot \sigma \mid \mathcal{B}[\![b]\!](s) = \textbf{tt}\} \cap \{s \cdot s \cdot \sigma \mid s \cdot \sigma \in \|\phi\|\} & \\ = & \{s \cdot \sigma \mid s \models b\} \cap \{s \cdot s \mid s \in \textbf{State}\}^{\frown} \{s \cdot \sigma \mid s \cdot \sigma \in \|\phi\|\} & \\ = & \|b\| \cap (\|Id\|^{\frown} \|\phi\|) & \\ = & \left\|b \wedge Id^{\frown}\phi\right\| & \end{array}$$

□

A.2. **Strongest Trace Formula.** The proof of Theorem 4.9—the semantics of the strongest trace formula of a program turns is identical to its trace semantics—requires an inner fixed-point induction for which it is advantageous to break down stf into a modal equation system mes [23] that preserves the structure of procedure declarations.

Definition A.6 (Modal Equation System)**.** Given a closed trace formula ϕ, the modal equation system $\mathsf{mes}(\phi)$ is an open trace formula together with a set of modal equations of the form $X_i = \phi_i$, inductively defined over ϕ as follows: $\mathsf{mes}(\phi)$ is just the homomorphism over the abstract syntax, except when $\phi = \mu X.\phi'$: In this case, $\mathsf{mes}(\phi) \stackrel{\mathsf{def}}{=} X$, and a new equation $X = \mathsf{mes}(\phi')$ is added.

The semantics of modal equation systems is defined as in the literature [23], from where we also take the semantic equivalence between ϕ and $\mathsf{mes}(\phi)$.

Example A.7. The modal equation system corresponding to the strongest trace formula in Example 4.7 is: $\mathsf{mes}(\mathsf{stf}(even())) = Id^{\frown}X_{even}$, where

$$X_{even} = \left((x = 0 \wedge Id^{\frown} Sb_y^1) \vee (x \neq 0 \wedge Id^{\frown} Sb_x^{x-1} {}^{\frown} Id^{\frown} X_{odd})\right)$$
$$X_{odd} = \left((x = 0 \wedge Id^{\frown} Sb_y^0) \vee (x \neq 0 \wedge Id^{\frown} Sb_x^{x-1} {}^{\frown} Id^{\frown} X_{even})\right)$$

Observe the structural similarity between the formula $Id^{\frown}X_{even}$ in the context of the defining equations for X_{even} and X_{odd}, and the statement $even()$ in the context of the procedure table T of Example 2.2.

To use the decomposition of a trace formula into a modal equation system, we need to define for each program S an open trace formula corresponding to $\mathsf{mes}(\mathsf{stf}(S))$. This transformation, called $\mathsf{stf}'(S)$, is defined exactly as $\mathsf{stf}(X, S)$ in Definition 4.6 (ignoring the parameter X), except for the case $S = m()$, which is defined as $\mathsf{stf}'(m()) = X_m$.

Lemma A.8. Let $\langle S, T\rangle$ be a **Rec** program and M the procedures declared in T. Let $\rho : M \to 2^{\mathbf{State}^+}$ be an arbitrary interpretation of the procedures $m \in M$, and let $\mathcal{V}_\rho : \mathsf{RVar} \to 2^{\mathbf{State}^+}$ be the (induced) valuation of the recursion variables defined by $\mathcal{V}_\rho(X_m) \stackrel{\mathsf{def}}{=} \rho(m)$. We have, for all statements S of **Rec**: $\left\|\mathsf{stf}'(S)\right\|_{\mathcal{V}_\rho} = \mathcal{S}^0_{tr}[\![S]\!]_\rho$

Proof. The proof proceeds by induction on the structure of S.
Case $S =$ **if** b **then** S_1 **else** S_2**.** By the induction hypothesis, $\|\mathsf{stf}(S_i)\|_{\mathcal{V}_\rho} = \mathcal{S}^0_{tr}[\![S_i]\!]_\rho$ for $i = 1, 2$. Using the obvious generalization of Proposition A.5 to open formulas, we have:

$$\begin{array}{rl} & \left\|\mathsf{stf}'(\textbf{if } b \textbf{ then } S_1 \textbf{ else } S_2)\right\|_{\mathcal{V}_\rho} \\ = & \left\|(b \wedge Id^\frown \mathsf{stf}'(S_1)) \vee (\neg b \wedge Id^\frown \mathsf{stf}'(S_2))\right\|_{\mathcal{V}_\rho} \\ = & \left\|(b \wedge Id^\frown \mathsf{stf}'(S_1))\right\|_{\mathcal{V}_\rho} \cup \left\|(\neg b \wedge Id^\frown \mathsf{stf}'(S_2))\right\|_{\mathcal{V}_\rho} \\ = & (\sharp \left\|\mathsf{stf}'(S_1)\right\|_{\mathcal{V}_\rho})|_b \cup (\sharp \left\|\mathsf{stf}'(S_2)\right\|_{\mathcal{V}_\rho})|_{\neg b} \\ = & (\sharp \mathcal{S}^0_{tr}[\![S_1]\!]_\rho)|_b \cup (\sharp \mathcal{S}^0_{tr}[\![S_2]\!]_\rho)|_{\neg b} \\ = & \mathcal{S}^0_{tr}[\![\textbf{if } b \textbf{ then } S_1 \textbf{ else } S_2]\!]_\rho \end{array}$$

Case $S = m()$**.** We have:

$$\begin{array}{rl} & \left\|\mathsf{stf}'(m())\right\|_{\mathcal{V}_\rho} \\ = & \|X_m\|_{\mathcal{V}_\rho} \\ = & \mathcal{V}_\rho(X_m) \\ = & \rho(m) \\ = & \mathcal{S}^0_{tr}[\![m()]\!]_\rho \end{array}$$

□

Proof of Theorem 4.9. Since $\mathsf{stf}'(S)$ and $\mathsf{stf}(X, S)$ are defined identically, except for the case $S = m()$, the proof is the same as for Lemma A.8, except for that case:
Case $S = m()$**.** We only sketch the proof here. We translate the formula $\mathsf{stf}(\varnothing, m())$ into a modal equation system $\mathsf{mes}(\mathsf{stf}(\varnothing, m()))$. This results in the formula X_m, defined in the context of a system of modal equations: for each fixed-point operator μX_i in $\mathsf{stf}(\varnothing, m())$, there is an equation $X_i = Id^\frown \mathsf{stf}'(S_i)$, whenever m_i is declared as $m_i\,\{S_i\}$ in T. Next, from the standard semantics of modal equation systems, and by Lemma A.8, it follows that the least solution $\mathcal{V}_0$ of the modal equation system is equal (on the names of the procedures called recursively by m) to the valuation $\mathcal{V}_{\rho_0}$ induced by the interpretation ρ_0 defined by the procedure table T. Finally, by Proposition A.4 and Lemma A.8, we have:

$$\begin{array}{rl} & \|\mathsf{stf}(\varnothing, m())\| \\ = & \|X_m\|_{\mathcal{V}_0} \\ = & \|X_m\|_{\mathcal{V}_{\rho_0}} \\ = & \mathcal{V}_{\rho_0}(X_m) \\ = & \rho_0(m) \\ = & \mathcal{S}^0_{tr}[\![m()]\!]_{\rho_0} \\ = & \mathcal{S}_{tr}[\![m()]\!] \end{array}$$

□

A.3. **Soundness of the Calculus.** For the proof we need to relate traces restricted by a condition b to trace formulas. In the following proposition, the intuition for $\|\neg b \vee \phi\|$ is that it ignores any trace, that is, it is trivially true for any trace, where b does not hold in the beginning.

Proposition A.9 (State formulas in judgments)**.** Let b be a Boolean expression, ϕ a trace formula. Then $(\mathcal{S}_{tr}[\![S]\!])|_b \subseteq \|\phi\| \quad \text{iff} \quad \mathcal{S}_{tr}[\![S]\!] \subseteq \|\neg b \vee \phi\|$.

Proof. "Only If" direction: Assume $(\mathcal{S}_{tr}[\![S]\!])|_b \subseteq \|\phi\|$ and there is a trace $s \cdot \sigma \in \mathcal{S}_{tr}[\![S]\!]$ that is not in $\|\neg b \vee \phi\| = \|\neg b\| \cup \|\phi\|$, hence, $s \cdot \sigma \notin \|\neg b\|$ and $s \cdot \sigma \notin \|\phi\|$. But $s \cdot \sigma \notin \|\neg b\|$ implies $s \cdot \sigma \in (\mathcal{S}_{tr}[\![S]\!])|_b \subseteq \|\phi\|$: contradiction.

"If" direction: Assume $\mathcal{S}_{tr}[\![S]\!] \subseteq \|\neg b \vee \phi\|$ and there is a trace $s \cdot \sigma \in (\mathcal{S}_{tr}[\![S]\!])|_b$ that is not in $\|\phi\|$. From the assumption and $(\mathcal{S}_{tr}[\![S]\!])|_b \subseteq \mathcal{S}_{tr}[\![S]\!]$ we obtain $s \cdot \sigma \in \|\neg b \vee \phi\|$. However, since $\mathcal{B}[\![b]\!](s) = \mathbf{tt}$ we must have in fact $s \cdot \sigma \in \|\phi\|$: contradiction. □

Proof of Theorem 5.6. The proof system is sound, since every rule of the system is locally sound, in the sense that its conclusion is valid whenever all its premises are valid. We shall prove local soundness of each rule. Without loss of generality, we ignore Γ in most cases.

Rule If.

Let $\mathcal{I}$ be an arbitrary interpretation. Using Proposition A.9, we have:

$$
\begin{array}{ll}
 & \models_{\mathcal{I}} \mathbf{skip}; S_1 : \neg b \vee \phi \text{ and } \models_{\mathcal{I}} \mathbf{skip}; S_2 : b \vee \phi \\
\Leftrightarrow & \mathcal{S}_{tr}[\![\mathbf{skip}; S_1]\!]_{\mathcal{I}} \subseteq \|\neg b \vee \phi\| \text{ and} \\
 & \mathcal{S}_{tr}[\![\mathbf{skip}; S_2]\!]_{\mathcal{I}} \subseteq \|b \vee \phi\| \\
\Leftrightarrow & \sharp\mathcal{S}_{tr}[\![S_1]\!]_{\mathcal{I}} \subseteq \|\neg b \vee \phi\| \text{ and } \sharp\mathcal{S}_{tr}[\![S_2]\!]_{\mathcal{I}} \subseteq \|b \vee \phi\| \\
\Leftrightarrow & (\sharp\mathcal{S}_{tr}[\![S_1]\!]_{\mathcal{I}})|_b \subseteq \|\phi\| \text{ and } (\sharp\mathcal{S}_{tr}[\![S_2]\!]_{\mathcal{I}})|_{\neg b} \subseteq \|\phi\| \\
\Leftrightarrow & ((\sharp\mathcal{S}_{tr}[\![S_1]\!]_{\mathcal{I}})|_b \cup (\sharp\mathcal{S}_{tr}[\![S_2]\!]_{\mathcal{I}})|_{\neg b}) \subseteq \|\phi\| \\
\Leftrightarrow & \mathcal{S}_{tr}[\![\mathbf{if}\ b\ \mathbf{then}\ S_1\ \mathbf{else}\ S_2]\!]_{\mathcal{I}} \subseteq \|\phi\| \\
\Leftrightarrow & \models_{\mathcal{I}} \mathbf{if}\ b\ \mathbf{then}\ S_1\ \mathbf{else}\ S_2 : \phi
\end{array}
$$

where we use that $\mathcal{S}_{tr}[\![\mathbf{skip}; S]\!]_{\mathcal{I}} = \sharp\mathcal{S}_{tr}[\![S]\!]_{\mathcal{I}}$, and therefore:

$$
\begin{array}{ll}
 & \models_{\mathcal{I}} \mathbf{skip}; S_1 : \neg b \vee \phi \text{ and } \models_{\mathcal{I}} \mathbf{skip}; S_2 : b \vee \phi \\
\Leftrightarrow & \models_{\mathcal{I}} \mathbf{if}\ b\ \mathbf{then}\ S_1\ \mathbf{else}\ S_2 : \phi
\end{array}
$$

Rule Call. For the proof of the rule we employ the principle of Fixed-Point Induction. To simplify the presentation, we assume there is only one procedure $m()$, declared as $m\{S_m\}$ in T. The general case follows from Bekič's Principle. The notation $\rho[m \mapsto \gamma]$ specifies the interpretation that is identical to ρ, except for $\rho(m) = \gamma$.

$$
\begin{array}{ll}
 & Y_m : \phi_m \models S_m[\mathbf{skip}; Y_m/m()] : \phi_m \\
\Leftrightarrow & \forall \mathcal{I}.\ (\mathcal{S}_{tr}[\![Y_m]\!]_{\mathcal{I}} \subseteq \|\phi_m\| \\
 & \qquad \Rightarrow \mathcal{S}_{tr}[\![S_m[\mathbf{skip}; Y_m/m()]]\!]_{\mathcal{I}} \subseteq \|\phi_m\|) \\
\Leftrightarrow & \forall \gamma.\ (\gamma \subseteq \|\phi_m\| \Rightarrow \mathcal{S}^0_{tr}[\![S_m]\!]_{\rho[m \mapsto \sharp\gamma]} \subseteq \|\phi_m\|) \\
\Leftrightarrow & \forall \gamma.\ (\sharp\gamma \subseteq \left\|Id^\frown \phi_m\right\| \Rightarrow \sharp\mathcal{S}^0_{tr}[\![S_m]\!]_{\rho[m \mapsto \sharp\gamma]} \subseteq \left\|Id^\frown \phi_m\right\|) \\
\Leftrightarrow & \forall \gamma.\ (\gamma \subseteq \left\|Id^\frown \phi_m\right\| \Rightarrow \sharp\mathcal{S}^0_{tr}[\![S_m]\!]_{\rho[m \mapsto \gamma]} \subseteq \left\|Id^\frown \phi_m\right\|) \\
\Rightarrow & \rho_0(m) \subseteq \left\|Id^\frown \phi_m\right\| \\
\Leftrightarrow & \mathcal{S}^0_{tr}[\![m()]\!]_{\rho_0} \subseteq \left\|Id^\frown \phi_m\right\| \\
\Leftrightarrow & \mathcal{S}_{tr}[\![m()]\!] \subseteq \left\|Id^\frown \phi_m\right\| \\
\Leftrightarrow & \models m() : Id^\frown \phi_m
\end{array}
$$

□

A.4. **Completeness of the Calculus.**

Proof of Theorem 5.7. The proof proceeds by induction on the structure of S. However, In the case of a call the statement does not necessarily get smaller, because the body of m is expanded. So we need to argue that the induction is well-founded. Indeed, the lexicographic order on $\langle N - |\Gamma|, |S| \rangle$ (obviously, the first component is never negative) always decreases. Since Γ is irrelevant for all cases except $S = m()$, we simplify the claim accordingly for these.

Case $S = \textbf{if } b \textbf{ then } S_1 \textbf{ else } S_2$**.** Assume by the induction hypothesis that $\vdash S_1 : \mathsf{stf}(S_1)$ and $\vdash S_2 : \mathsf{stf}(S_2)$ can be proven. We also use that $\phi \models (p \vee (\neg p \wedge \phi))$ is a valid consequence. We have:

$$
\begin{array}{ll}
 & \vdash \textbf{if } b \textbf{ then } S_1 \textbf{ else } S_2 : \mathsf{stf}(\textbf{if } b \textbf{ then } S_1 \textbf{ else } S_2) \\
\Leftrightarrow & \vdash \textbf{if } b \textbf{ then } S_1 \textbf{ else } S_2 : (b \wedge Id^\frown \mathsf{stf}(S_1)) \vee \\
 & \qquad\qquad\qquad\qquad (\neg b \wedge Id^\frown \mathsf{stf}(S_2)) \\
\Leftarrow & \{\text{By rule (If) combined with rule (Or)}\} \\
 & \vdash \textbf{skip}; S_1 : \neg b \vee (b \wedge Id^\frown \mathsf{stf}(S_1)) \text{ and} \\
 & \vdash \textbf{skip}; S_2 : b \vee (\neg b \wedge Id^\frown \mathsf{stf}(S_2)) \\
\Leftarrow & \{\text{By rule (Cons)}\} \\
 & \vdash \textbf{skip}; S_1 : Id^\frown \mathsf{stf}(S_1) \text{ and } \vdash \textbf{skip}; S_2 : Id^\frown \mathsf{stf}(S_2) \\
\Leftrightarrow & \{\text{By rule (Seq) combined with rule (Skip) and} \\
 & \text{the induction hypothesis}\} \\
 & \mathsf{true}
\end{array}
$$

Case $S = m()$**.** There are two subcases: either $m \in \{m_1, \ldots, m_n\}$ or not. In the first case, $S[\textbf{skip}; Y_{m_1}/m_1(), \ldots, \textbf{skip}; Y_{m_n}/m_n()] = \textbf{skip}; Y_m$

In the second case, $S[\textbf{skip}; Y_{m_1}/m_1(), \ldots, \textbf{skip}; Y_{m_n}/m_n()] = m()$. In both cases, we have $\mathsf{stf}(m()) = Id^\frown \phi_m$.

Subcase skip$; Y_m$

We have to prove the judgment $\Gamma \vdash \textbf{skip}; Y_m : Id^\frown \phi_m$. Using rules (Seq) and (Skip), this reduces to $\Gamma \vdash Y_m : \phi_m$. Because of the assumption $m \in \{m_1, \ldots, m_n\}$ we have $Y_m : \phi_m \in \Gamma$, so the proof is finished.

Subcase $m()$

We have to prove the judgment $\Gamma \vdash m() : Id^\frown \phi_m$. Rule (Call) is applicable due to assumption $m \notin \{m_1, \ldots, m_n\}$. Let

$$S'_m \stackrel{\mathsf{def}}{=} S_m[\textbf{skip}; Y_m/m(), \textbf{skip}; Y_{m_1}/m_1(), \ldots, \textbf{skip}; Y_{m_n}/m_n()]$$

The obtained premise yields the new claim to prove:

$$\Gamma \cup \{Y_m : \phi_m\} \vdash S'_m : \mu X_m.\, \mathsf{stf}(\{m\}, S_m)$$

We apply rule (Unfold) on the right and obtain:

$$\Gamma \cup \{Y_m : \phi_m\} \vdash S'_m : \mathsf{stf}(\{m\}, S_m)[\phi_m / X_m]$$

By induction and by the definition of $\mathsf{stf}(S)$ we finish the proof up to subgoals of the form (for some $m' \neq m$):

$$\Gamma \cup \{Y_m : \phi_m\} \vdash m'() : \mathsf{stf}(\{m\}, m')[\phi_m / X_m]$$

Unfortunately, the structure of the formula on the right does not conform to $\mathsf{stf}(m')$ as required. But, observing that $\|\mathsf{stf}(S)\| = \|\mathsf{stf}(\overline{X}, S)\|$, as well as soundness of unfolding, we can use Cons to obtain:

$$\Gamma \cup \{Y_m : \phi_m\} \vdash m'() : \mathsf{stf}(m')$$

This follows from the induction hypothesis, because of $N - |\Gamma| > N - |\Gamma \cup \{Y_m : \phi_m\}|$.

□

A.5. **Canonical Programs.**

Proof of Proposition 6.5. We need to show that for any call $m_X()$ in $\mathsf{can}(\phi)$ there is exactly one declaration of m_X in T_ϕ. This is a straightforward structural induction. □

Lemma A.10. Let ϕ be an open trace formula not containing fixed-point binders ($\mu X.$), and let $\mathsf{can}(\phi) = \langle S_\phi, T_\phi \rangle$. Then, we have: $\mathcal{S}^0_{tr}[\![S_\phi]\!]_\rho \mathrel{\widetilde{=}} \|\phi\|_{\mathcal{V}_\rho}$ for all interpretations $\rho : M \to 2^{\mathbf{State}^+}$ of the procedures M declared in T_ϕ, and (induced) valuations $\mathcal{V}_\rho : \mathsf{RVar} \to 2^{\mathbf{State}^+}$ defined by $\mathcal{V}_\rho(X_m) \stackrel{\mathsf{def}}{=} \rho(m)$.

Proof. We proceed by structural induction on ϕ.
Case $\phi = p \wedge \psi$
By the induction hypothesis, $\mathcal{S}^0_{tr}[\![S_\psi]\!]_\rho \mathrel{\widetilde{=}} \|\psi\|_{\mathcal{V}_\rho}$.

$$
\begin{aligned}
&\mathcal{S}^0_{tr}[\![\textbf{if } p \textbf{ then } S_\psi \textbf{ else diverge}]\!]_\rho \\
&\quad = (\sharp \mathcal{S}^0_{tr}[\![S_\psi]\!]_\rho)|_p \cup (\sharp \mathcal{S}^0_{tr}[\![\textbf{diverge}]\!]_\rho)|_{\neg p} \\
&\quad = (\sharp \mathcal{S}^0_{tr}[\![S_\psi]\!]_\rho)|_p \qquad \text{(Proposition 6.4)} \\
&\quad = \mathbf{State}^+|_p \cap \sharp \mathcal{S}^0_{tr}[\![S_\psi]\!]_\rho \mathrel{\widetilde{=}} \mathbf{State}^+|_p \cap \mathcal{S}^0_{tr}[\![S_\psi]\!]_\rho \\
&\quad \mathrel{\widetilde{=}} \mathbf{State}^+|_p \cap \|\psi\|_{\mathcal{V}_\rho} \quad \text{(Induction hypothesis)} \\
&\quad = \|p \wedge \psi\|_{\mathcal{V}_\rho}
\end{aligned}
$$

Case $\phi = \phi_1 \vee \phi_2$
By the induction hypothesis, we have that $\mathcal{S}^0_{tr}[\![S_{\phi_1}]\!]_\rho \mathrel{\widetilde{=}} \|\phi_1\|_{\mathcal{V}_\rho}$ and $\mathcal{S}^0_{tr}[\![S_{\phi_2}]\!]_\rho \mathrel{\widetilde{=}} \|\phi_2\|_{\mathcal{V}_\rho}$. Therefore,

$$
\begin{aligned}
\mathcal{S}^0_{tr}[\![\textbf{if } * \textbf{ then } S_{\phi_1} \textbf{ else } S_{\phi_2}]\!]_\rho &= \sharp \mathcal{S}^0_{tr}[\![S_{\phi_1}]\!]_\rho \cup \sharp \mathcal{S}^0_{tr}[\![S_{\phi_2}]\!]_\rho \\
\mathrel{\widetilde{=}}\ \mathcal{S}^0_{tr}[\![S_{\phi_1}]\!]_\rho \cup \mathcal{S}^0_{tr}[\![S_{\phi_2}]\!]_\rho &\mathrel{\widetilde{=}} \|\phi_1\|_{\mathcal{V}_\rho} \cup \|\phi_2\|_{\mathcal{V}_\rho} \quad \text{(Ind. hyp.)} \\
&= \|\phi_1 \vee \phi_2\|_{\mathcal{V}_\rho}
\end{aligned}
$$

Case $\phi = X$
We have:

$$\mathcal{S}^0_{tr}[\![m_X()]\!]_\rho = \rho(m_X) = \mathcal{V}_\rho(X) = \|X\|_{\mathcal{V}_\rho}$$

□

Proof of Theorem 6.7. By structural induction on ϕ. We only sketch the proof idea here. An argument can be made similar to the one in the last case of the proof of Theorem 4.9, by referring to the modal equation system corresponding to ϕ, and using Lemma A.10 to generalise the treatment to open formulas ϕ and statements that contain calls to procedures not declared in T_ϕ. Proposition 6.5 ensures that the program on the left is well-defined. Compositionality of the proof is justified by Bekič's Principle. □

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